

JalDrishti (CoastalEye)

Comprehensive Technical Architecture, System Manifest & Engineering Documentation

Version: 2.0 (Production Ready)	Framework: React 19 + FastAPI	Database: MongoDB Atlas + Motor	Status: Fully Verified (100% Tests Pass)
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1. Executive Overview & System Architecture

JalDrishti (CoastalEye) is a unified, real-time AI-powered water hazard reporting, municipal dispatch, and emergency SOS relief platform. The solution integrates a dual-portal architecture comprising a public **Citizen Portal** and a specialized **Government Disaster Management Portal**. The system features multimodal AI verification via Google Gemini, geospatial hotspot clustering, dynamic municipal department assignment, real-time lifecycle tracking, and an automated 24-hour database retention and auto-purge policy for concluded incidents.

2. Full Technology Stack

Layer	Technologies	Key Features & Responsibilities
Frontend	React 19, TypeScript, Vite, Tailwind CSS, Leaflet, Lucide Icons	Citizen hazard reporting, SOS rescue relief form, real-time report tracker with stage stepper, government review desk, interactive geospatial maps, dynamic profile manager.
Backend	Python 3.12, FastAPI, Uvicorn, AsyncIO	High-performance asynchronous REST API handling report ingestion, city/state filtering, assignment workflows, rescue team dispatch, and JWT session handling.
Database	MongoDB Atlas, Motor (Async ODM), PyMongo	Document storage with schema-flexible hazard reports, relief missions, user profiles, TTL indices for 24-hour auto-purge, and spatial coordinates.
AI & ML	Google Gemini API, Spatial Clustering Engine	Multimodal image analysis, automated severity rating, categorization, and Haversine distance spatial grouping for incident hotspot identification.
Auth & Security	JWT (HMAC-SHA256), Passlib (Bcrypt), OAuth2	Role-based access control (Citizen vs. Government), dynamic profile updates, token re-issuance, and protected API endpoints.

3. Backend File Manifest & Component Roles

File Path	Component Role & Functional Changes
Backend/app/reports/service.py	Core report business logic. Implements report creation, department assignment (assignReport), status transitions (updateReportStatus), rejection handling with reason recording, timeline event deduplication, and cleanup_expired_reports() for 24-hour MongoDB auto-deletion.
Backend/app/reports/routes.py	FastAPI REST router for reports. Exposes endpoints for report creation (POST /reports/), filtered retrieval (GET /reports/), tracking by public ID (GET /reports/{id}), department assignment, and status updates.
Backend/app/relief/service.py	Emergency SOS relief operations. Manages NDRF/SDRF rescue team assignments, resource allocations, status transitions, deleteReliefRequest() for permanent document removal, and 24-hour expiration tracking.

Backend/app/relief/routes.py	FastAPI REST router for rescue requests. Exposes POST /relief/, GET /relief/, status update routes, and DELETE /relief/{requestId} for permanent deletion.
Backend/app/auth/service.py	Authentication and profile management. Removed hardcoded defaults ('Disaster Officer Verma'). Implements update_user_profile to persist dynamic name, phone, department, and designation edits in MongoDB.
Backend/app/auth/routes.py	Auth API router. Exposes POST /auth/login, GET /auth/me, and PATCH /auth/profile to update user attributes and issue refreshed JWT tokens.
Backend/app/database.py	Motor asynchronous MongoDB client configuration with TLS/SSL certificate verification (certifi).
Backend/app/models/hotspot_clustering.py	Spatial clustering algorithm using Haversine distance to aggregate proximate reports into high-risk hazard zones.

4. Frontend File Manifest & Component Roles

File Path	Component Role & Functional Changes
src/pages/Citizen/TrackReport.tsx	Citizen live status tracking page. Fixed stage index mapping so Assigned and In Progress stages highlight with active pulses and green checkmarks. Renders red Rejection Banner with official reason, 24-Hour Auto-Purge Countdown Timer , and clean chronological action history.
src/pages/Citizen/RescueRelief.tsx	Citizen emergency SOS relief portal. Allows citizens to log rescue requests with affected people count and urgency level. Supports case-insensitive live tracking for NDRF/SDRF deployment.
src/pages/Citizen/Profile.tsx	Citizen profile view allowing live updating of full name, phone number, and address with instant JWT and state synchronization.
src/pages/Government/Dashboard.tsx	Executive disaster control dashboard. Removed hardcoded 'Kanpur' defaults. Dynamically aggregates city/state specific stats (Pending, Active, Resolved, Rejected, Critical).
src/pages/Government/ReviewReports.tsx	Government desk report review queue. Features live status tabs, city filtering, priority filters, and one-click access to the detail drawer.
src/components/report/GovernmentReportDetailDrawer.tsx	Report inspection drawer. Fixed JSX parsing issue. Added department assignment dropdown (Drainage, Jal Nigam, PWD, NDRF) and status controls. Triggers rejection modal upon reject.
src/components/report/RejectReasonModal.tsx	Mandatory rejection modal. Captures officer reason with preset tags ('Duplicate Submission', 'False Alarm', 'Not a Water Hazard') and submits to backend.
src/pages/Government/RescueRequests.tsx	Government rescue mission management. Added Mark as Resolved button to complete operations and Remove button with confirmation modal to permanently delete records from MongoDB.
src/pages/Government/Profile.tsx	Government officer profile management. Allows officers to rename their profile, designation, and department. Immediately updates Topbar, token, and assignment audit logs.
src/services/reportAdapters.ts	Type-safe adapters mapping MongoDB documents to frontend interfaces (TrackedReportView, WaterReport). Passes assignedDepartment, concludedAt, and expiresAt.
src/context/AuthContext.tsx	Authentication state provider. Implements updateUser() to synchronize user updates live across the React tree and localStorage.

5. Summary of Core Issues Resolved & Enhancements Built

- **Dynamic Profile Renaming:** Removed all hardcoded names ('Disaster Officer Verma'). Updating an officer profile in /government/profile instantly syncs the Topbar, Sidebar, auth token, and future assignment audit entries.
- **Multi-Jurisdiction City Filter:** Removed hardcoded Kanpur defaults. Dashboard and reports dynamically auto-detect location or adjust according to selected city/state filters.
- **Rejection Reason Modal & Unlisting:** Rejection requires a mandatory reason, stores it in verification.officerNotes, unlists the report from active queues, and displays the reason to citizens.
- **Rescue Mission Resolution & Permanent Deletion:** Added both 'Mark as Resolved' and 'Remove' buttons for rescue requests to conclude or permanently purge documents from MongoDB.
- **Citizen Track Report Stage Synchronization:** Added 'assigned' to STATUS_ORDER so that Assigned and In Progress stages highlight with pulsing active badges and checkmarks.
- **Timeline Deduplication & Consolidation:** Cleaned up repetitive status entries (e.g. repeated 'Status updated to Assigned') on both backend and frontend.
- **24-Hour Database Retention & Auto-Purge Policy:** Concluded reports (Resolved / Rejected) automatically receive an expiresAt timestamp and are permanently deleted from MongoDB after 24 hours, with a live countdown timer displayed to citizens.

6. Verification & Test Suite Results

Test Suite / Command	Scope Tested	Result
npm run build	Full TypeScript compilation & Vite production bundle generation.	100% PASS (0 Errors)
test_profile_rename.py	Dynamic profile rename, token refresh, and Topbar sync.	100% PASS
test_rescue_resolve_delete.py	Rescue mission resolution and permanent MongoDB deletion.	100% PASS
test_features_report.py	Report lifecycle (Submitted -> Assigned -> In Progress -> Resolved / Rejected).	100% PASS
test_24h_timer_and_stages.py	Citizen Track stage indexing, 24-hour expiration timer & auto-purge.	100% PASS